

15. Fill in the two blank lines below to indicate what output you expect to see displayed in response to the previous bash commands in this session.

```
student@cs400:~/midterm$ ls
fileA fileB folder1/ folder2/
student@cs400:~/midterm$ ls folder1
fileC
student@cs400:~/midterm$ ls folder2
fileD
student@cs400:~/midterm$ mv folder2/fileD folder1/fileE
student@cs400:~/midterm$ cp folder1/fileC ../fileF
student@cs400:~/midterm$ rm fileA
student@cs400:~/midterm$ cd folder1
student@cs400:~/midterm/folder1$ ls
```

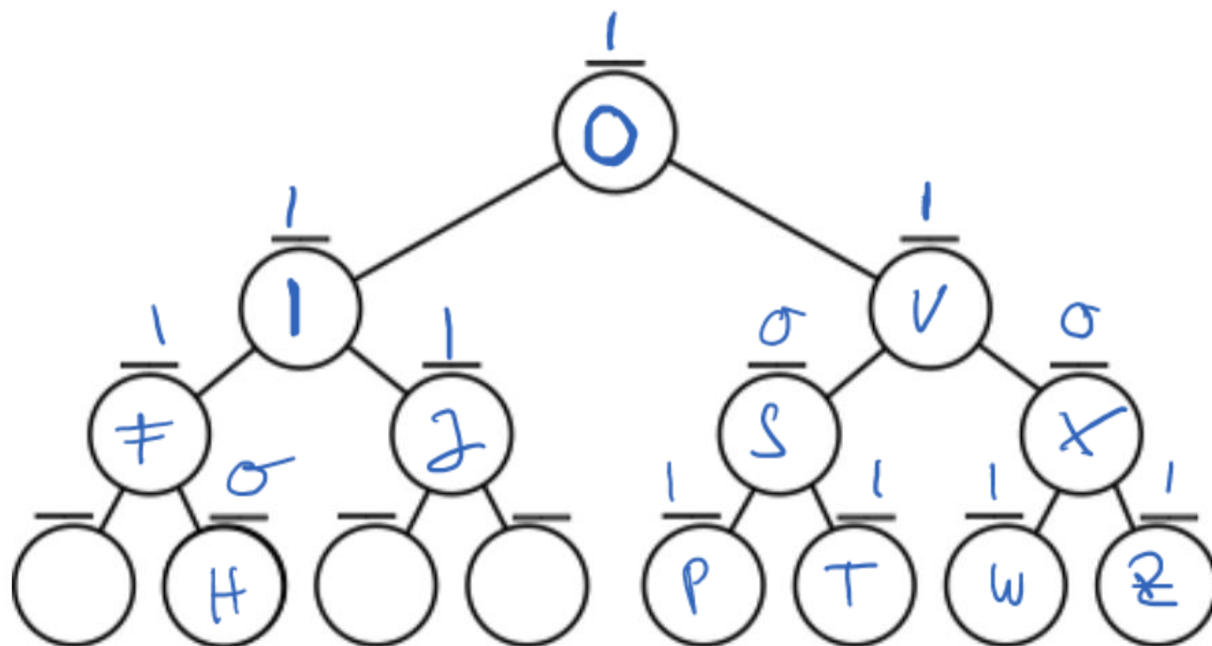
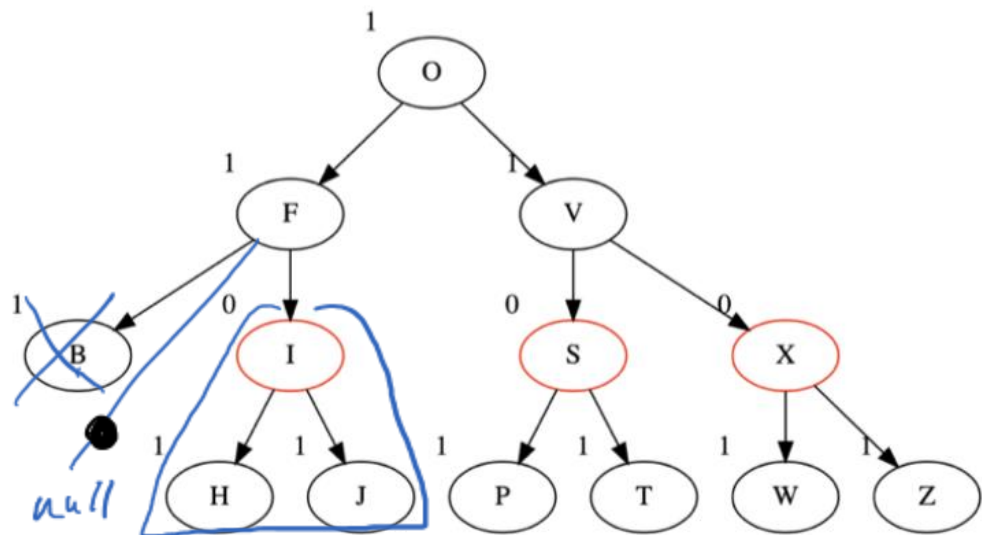
fileC, fileE

```
student@cs400:~/midterm/folder1$ ls ..
```

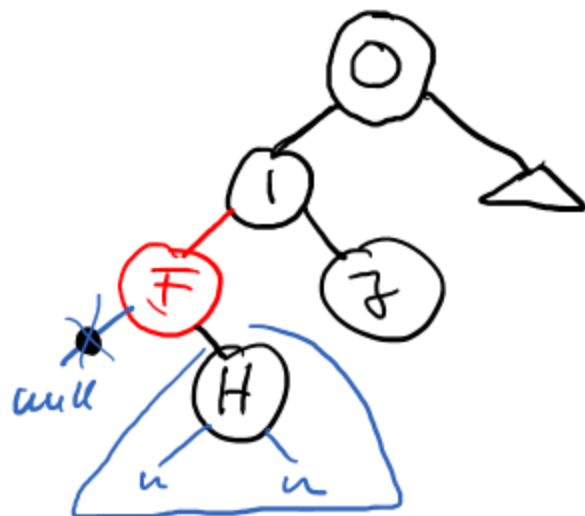
fileB, folder1, folder2



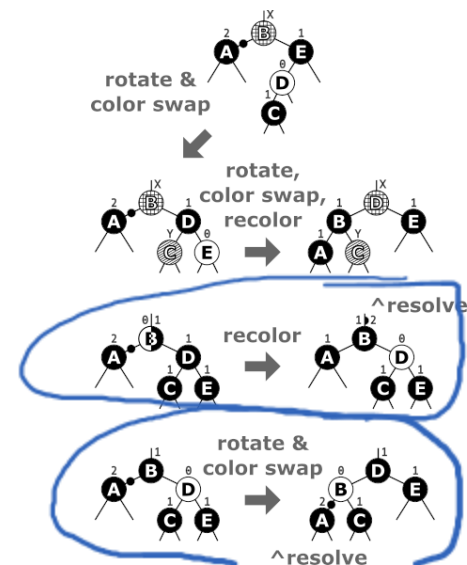
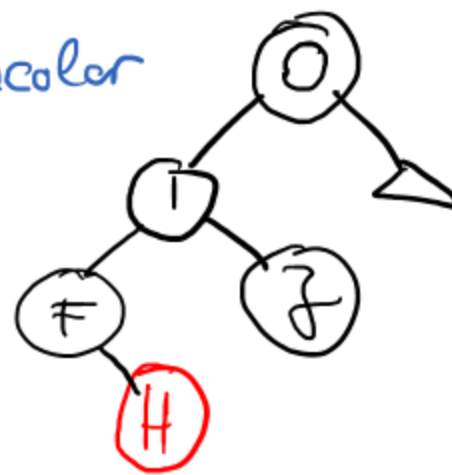
14. Remove the letter B from the following red-black tree, and then draw the resulting tree into the provided blanks.



1) rotate I, F
 + color swap I, F



2) recolor



13. Complete the final case of the rotation implementation below, to rotate the specified child and parent nodes. Note that this is simpler than P102 both 1) because nodes do not have parent references that need updating, and 2) because you are only implementing this one very specific case: where a child is the left child of its parent, and that parent is a left child of its parent, and all three references are non-null.

```
protected static class Node<T> {
    public Node<T> leftChild;
    public Node<T> rightChild;
    public boolean isLeftChild() { /* implementation details omitted */ }
}
```

```
public void rotate(Node<T> child, Node<T> parent, Node<T> grandParent) {
```

```
    // only implement the following case for this method:
```

```
    if( child.isLeftChild() && parent.isLeftChild() ) {
```

```
    // note that child, parent, and grandparent are all non-null in this case
```

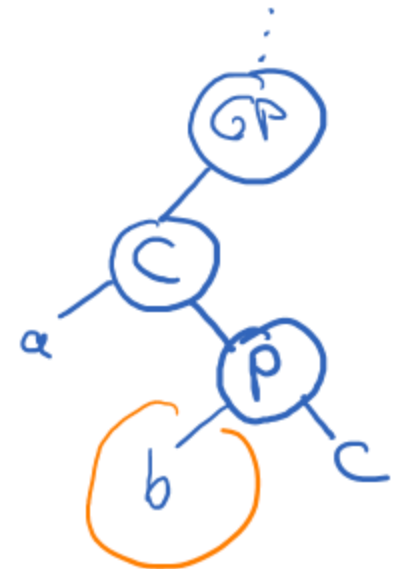
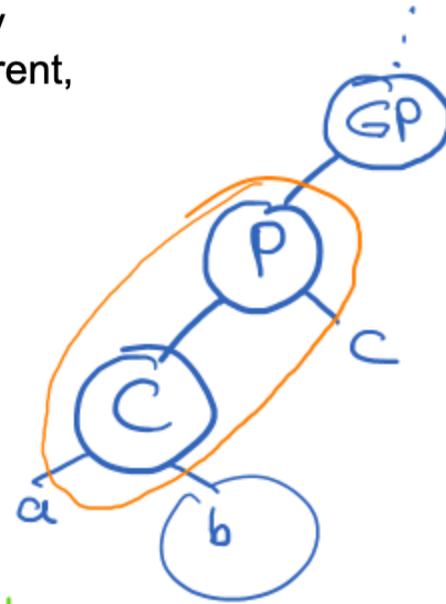
grandParent.leftChild = child;

parent.leftChild = child.rightChild;

child.rightChild = parent;

```
}
```

```
}
```



12. Which of the following lambda expressions can be used to implement the Comparable<T> interface?

```
public interface Comparable<T> {  
    public int compareTo(T other);  
}
```

A. (x) -> System.out.println(-1) ✗

B. (x) -> x ✗

C. () -> 16 ✗

D. (x) -> 3 ✓

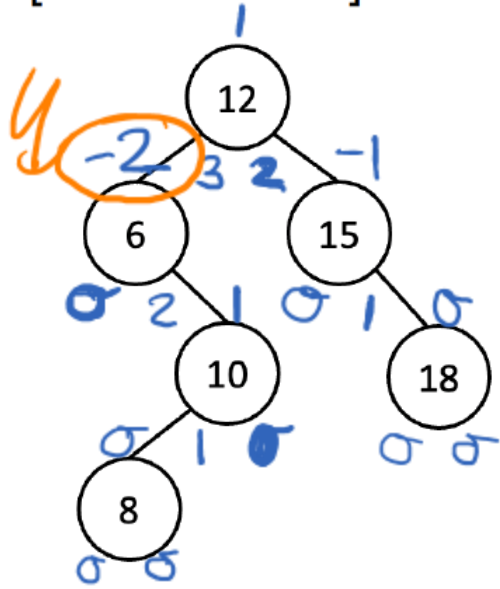
11. [True or False] The JUnit test runner will report two passing tests from the following Test class.

```
public class Tester {  
    @Test public void test1() {}  
    @Test public void test2() { Assertions.assertTrue(true); }  
}
```

run
passing
run
passing

- A. True
- B. False

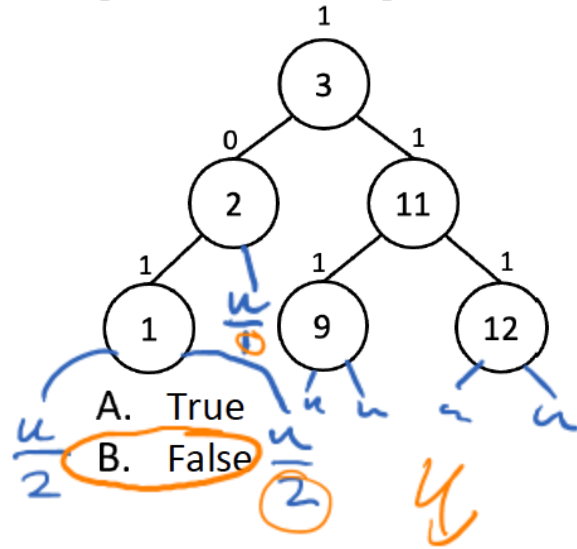
10. [True or False] The following tree is a balanced AVL tree.



A. True

B. False

9. [True of False] The following tree is a valid Red-Black Tree.



8. Consider running "make all" in the directory below. Which of the following files do you expect to see created as a result of running this command?

```
[student@exam] (1)$ ls -l
```

```
total 1
```

```
-rw-rw-r-- 1 student student 69 Mar 18 17:30 Makefile
```

```
-rw-rw-r-- 1 student student 0 Mar 18 17:31 X
```

```
-rw-rw-r-- 1 student student 0 Mar 18 17:32 Y
```

```
-rw-rw-r-- 1 student student 0 Mar 18 17:33 Z
```

```
[student@exam] (2)$ cat Makefile
```

```
all: W X Z
```

```
ls
```

→ W:

```
touch A
```

creates file A

→ X: Y

```
touch B
```

creates file B

→ Z: Y

```
touch C
```

A. A, B, C

B. A, B

C. A, C

D. B, C

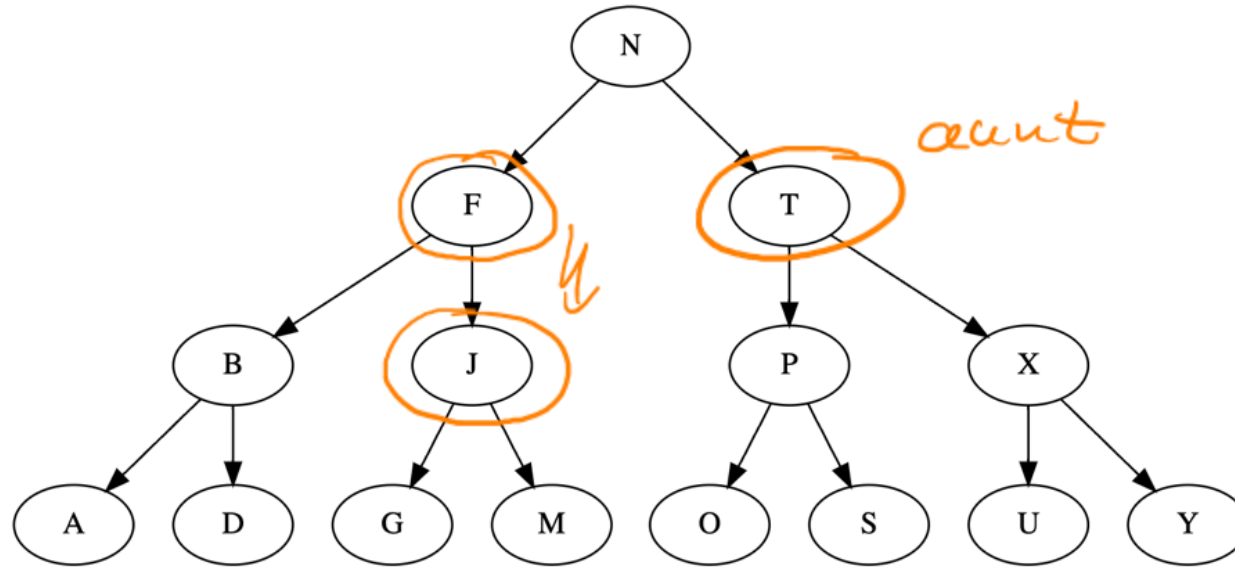
E. Only B

7. Which of the following tests is best categorized as a unit test?

- A. A test that runs a backend for Project 1 with an AVL tree implementation to check if the backend returns the expected data from a dataset stored in the tree.
- ☒ B. A test that passes two unrelated nodes into the rotate method of the BinarySearchTree of Project 1 to test if the method throws an exception.
- C. A test that runs a frontend with a backend for Project 1 to test if the frontend lists the songs for a given range from the dataset read in by the backend.

6. Suppose that J is a red node with a red parent in the following tree, and that outside of this one violation the rest of the following tree conforms to all Red-Black Tree properties (the colors are just omitted from this diagram). Which of the following nodes would you need to know the color of to determine the appropriate case and algorithm to begin resolving this violation?

- A. T
- B. M
- C. N
- D. G
- E. B



5. Which of the following commands can be used to check which files have been added to staging, and are ready to be added to the next commit?

- ~~✗~~A. git log
- ~~✗~~B. git commit
- C. git status
- ~~✗~~D. git add
- ~~✗~~E. git checkout

4. How many basic (single) rotations are used throughout the removal of the value K from the red-black tree depicted to the right?

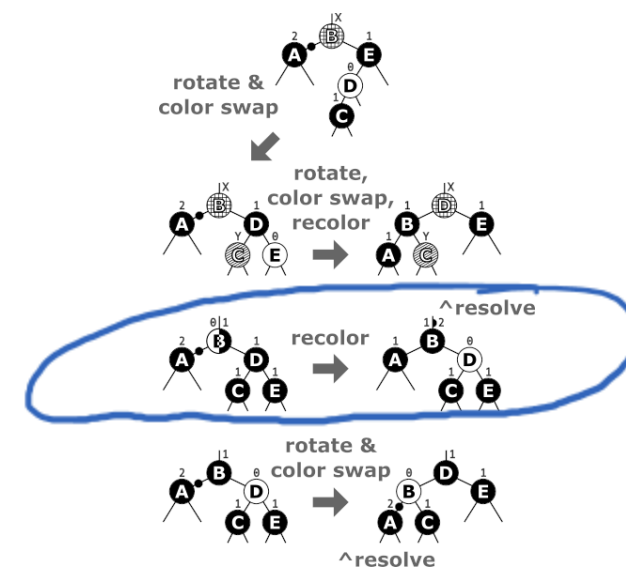
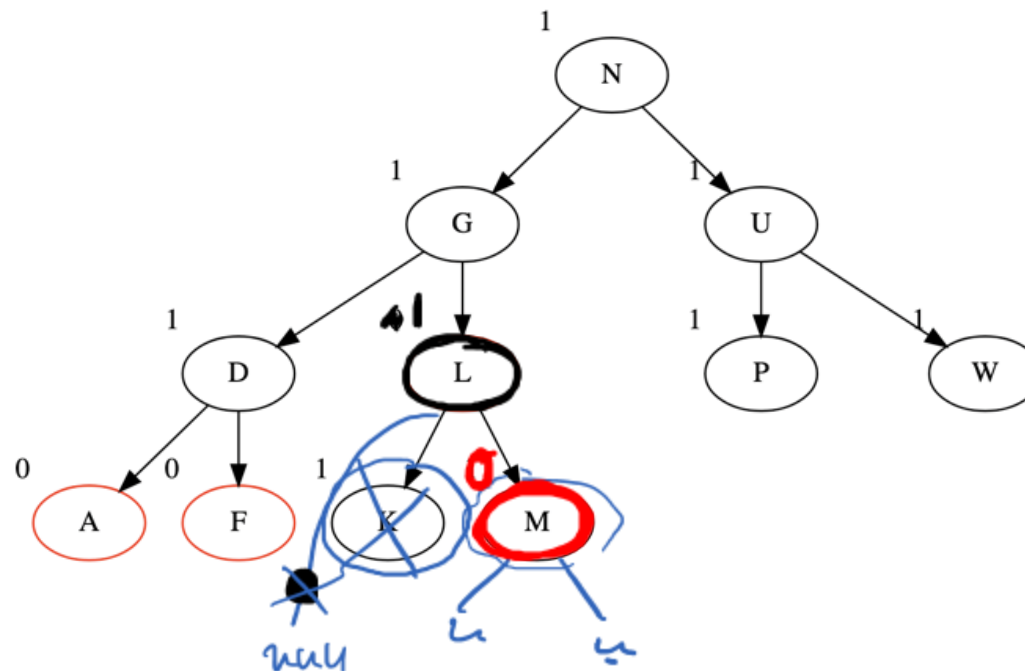
A. 0

B. 1

C. 2

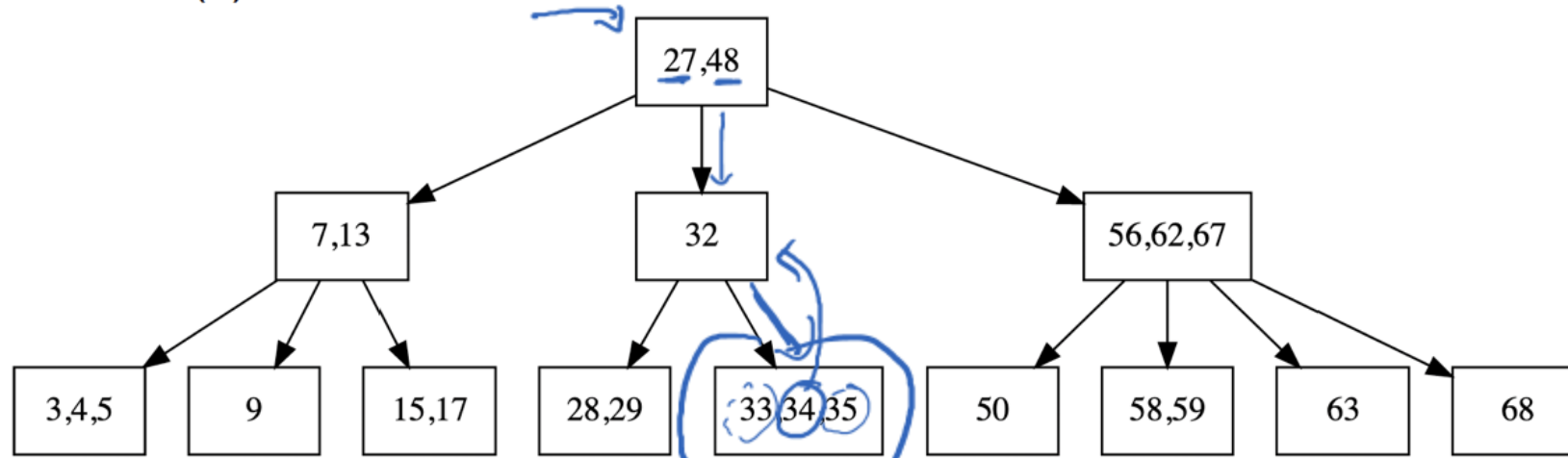
D. 3

E. 4

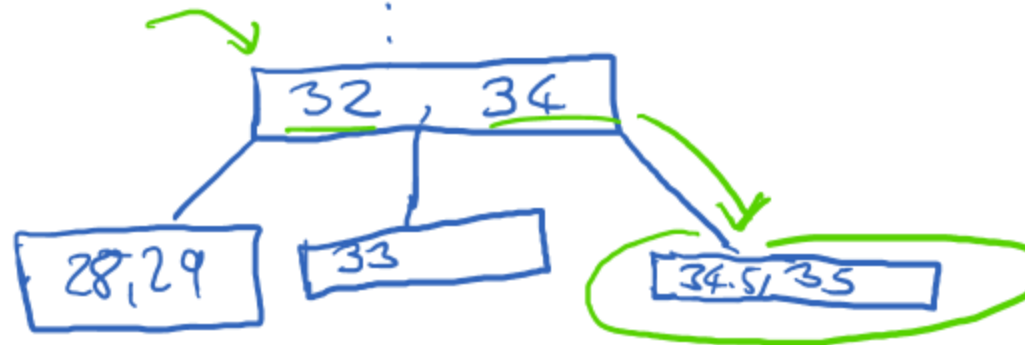


3. Trace through inserting the number 34.5 into the 2-3-4 tree below. After this insertion, which value(s) below will 34.5 share a node with?

- A. 32
- B. 34
- C. 35
- D. 32,34
- E. 27,48



⇒ Split



2. Which of the following commands could potentially change your working directory to be a sibling directory of the working directory?

A. cd a/./

B. cd b/././

C. cd ../c/

D. cd .././d

E. cd ../e/./

1. Consider rotating the nodes containing 6 and 10 in the binary search tree that is diagramed to the right. Which of the following statements will be true about the resulting tree?

- ☒ A. 8 will be a **LEFT** child, and the height of the resulting tree will be **SMALLER**
- ☒ B. 8 will be a **LEFT** child, and the height of the resulting tree will be **THE SAME**
- ☒ C. 8 will be a **LEFT** child, and the height of the resulting tree will be **LARGER**
- ☒ D. 8 will be a **RIGHT** child, and the height of the resulting tree will be **SMALLER**
- ☒ E. 8 will be a **RIGHT** child, and the height of the resulting tree will be **THE SAME**

